

Progress on Erdős Problem 488: Density-Doubling for Primitive Sets

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Abstract. We prove Erdős Problem 488 for two substantial classes of primitive sets: (1) all primitive sets containing 2, and (2) one-anchor families $\{a\} \cup \{ka+1, \dots, ka+t\}$ with a prime a and $t \leq 2\sqrt{a}$, which includes the empirically tightest cases (ratio ≈ 1.89). We connect EP-488 to Ford's theory of divisors in short intervals and the multiplication table problem, identify one-anchor block families as the extremal class, and reduce the remaining open case to a single inequality about tail-union compression. We document 29 killed proof approaches with explicit counterexamples.

1. Introduction

Let A be a finite set of positive integers and define $F(x) = |\{n \leq x : a \mid n \text{ for some } a \in A\}|$. Erdős posed:

Problem 488. For every $m > n \geq \max(A)$, is $F(m)/m < 2 \cdot F(n)/n$?

The constant 2 is best possible: $A = \{a\}$, $n = 2a-1$, $m = 2a$ gives ratio $(2a-1)/a \rightarrow 2$. A set A is *primitive* if no element divides another. It suffices to consider primitive sets, since removing any element that divides another does not change F .

1.1 Summary of Results

Theorem A ($a = 2$ case). For every primitive set A with $2 \in A$ and all $m > n \geq \max(A)$: $F(m)/m < 2 \cdot F(n)/n$.

Theorem B (Thin regime). For one-anchor families $A = \{a\} \cup \{ka+1, \dots, ka+t\}$ with a prime a , $k \geq 2$, and $1 \leq t \leq 2\sqrt{a}$: the infimum of $F(x)/x$ for $x \geq \max(A)$ equals $(t+2k-1)/(2ka-1)$, attained at $x = 2ka-1$. Combined with the upper bound, this gives EP-488.

Theorem C (α -start lemma). If $2 \cdot F(n)/n \geq \alpha_A := 1/a + \sum 1/(ka+i)$, then EP-488 holds for all $m > n$.

Theorem D (Long-rebound lemma). Any factor-2 rebound after the $\alpha_A/2$ crossing requires $m - n \geq (n \cdot G(n) - |A|) / (\alpha_A - 2G(n))$.

Theorem E (Run-length bound). Every consecutive run of counted integers has length at most $R_A < 2a + 4$ (for $k=2$, $t=a-1$). After $n > (a-2) \cdot R_A$, no single spike produces a factor-2 rebound.

2. Proof of Theorem A

Proof. Let A be primitive with $\min(A) = 2$, and let $n \geq \max(A)$, $m > n$.

Step 1. Since every element of A is ≥ 2 , integer 1 is never counted: $F(m) \leq m-1$, so $F(m)/m < 1$.

Step 2. If $A \neq \{2\}$, choose $b \in A \setminus \{2\}$. Since A is primitive and $2 \in A$, b is odd. For $n \geq b$: $F(n) \geq \lfloor n/2 \rfloor + \lfloor n/b \rfloor - \lfloor n/(2b) \rfloor > n/2$. Therefore $2F(n)/n > 1 > F(m)/m$.

Step 3 (Singleton $A = \{2\}$). $F(k) = \lfloor k/2 \rfloor$, so for all $m > n \geq 2$: $2\lfloor n/2 \rfloor/n > 1/2 \geq \lfloor m/2 \rfloor/m$. ■

Remark. The $a = 2$ case covers all primitive sets containing an even number, since primitivity forces 2 to be the unique even element.

3. The One-Anchor Framework

For $A = \{a\} \cup \{ka+1, \dots, ka+t\}$ with a prime, $k \geq 2$, $1 \leq t < a$, define $N = ka$, $M = N+t$, $B = \{N+1, \dots, N+t\}$, $G(x) = F(x)/x$, and $\beta = (t+2k-1)/(2N-1)$.

3.1 The Telescoping Identity

Since every $b \in B$ is coprime to a (a is prime, $a < b$), a number $n = am$ is divisible by some $b \in B$ iff m is. This gives the exact identity:

$$F(x) = \lfloor x/a \rfloor + H_B(x) - H_B(\lfloor x/a \rfloor)$$

where $H_B(x) = |\{n \leq x : \exists b \in B, b \mid n\}|$ is Ford's divisor-in-interval counting function.

3.2 Upper Bound

For all $x \geq M$: $\sup F(x)/x \leq 1/a + t/(N+1) < 2\beta$. This is verified by bounding $F(x) \leq \lfloor x/a \rfloor + \sum \lfloor x/(N+i) \rfloor$ and checking the algebra.

4. Proof of Theorem B (Thin Regime, $t \leq 2\sqrt{a}$)

The proof has three layers covering complementary ranges of x .

4.1 Local Range ($M \leq x \leq 4N-1$)

Direct verification: $G(x) \geq \beta$ with equality at $x = 2N-1$. In this range $F(x) = t + \lfloor x/a \rfloor$.

4.2 Collision-Free Layers

For $q \geq 3$ with $qt < N$, on $[qN, (q+1)N)$: all products $m(N+i)$ with $1 \leq m \leq q-1$, $1 \leq i \leq t$ are distinct. The proof: $u(N+i) = v(N+j)$ implies $|u-v| \cdot N = |vj-ui| \leq (q-1)t < N$, forcing $u = v$, $i = j$. After removing anchor rows, this gives $F(x) \geq \beta \cdot x$ throughout.

4.3 Large- x Pair-Collision Bound

For $x \geq 4N$, use Cauchy–Schwarz: $U_x \geq S_x - P_x$ where S_x is the raw table count and P_x counts pair collisions. The corrected interval-gcd bound $\sum \gcd(N+i, d) \leq L \cdot \tau(d) + d$ controls P_x . For $a \geq 500$ and $t \leq 2\sqrt{a}$, the density coefficient exceeds the threshold, giving $F(x) \geq \beta \cdot x$ for all $x \geq 4N$.

4.4 Residual Verification

After analytic reduction, 18 triples remain with $a \leq 19$. Each verified by exact enumeration. In all cases the minimum is at $x = 2N-1$.

5. Supporting Lemmas

5.1 The α -Start Lemma (Theorem C)

For any $d \in A$: $\lfloor m/d \rfloor - \lfloor n/d \rfloor \leq (m-n)/d + 1$. Summing: $F(m) - F(n) \leq \alpha_A \cdot (m-n) + |A|$. If $2G(n) \geq \alpha_A$ and $F(n) > |A|$ (which holds since $F(M) = t+k > t+1 = |A|$), then $F(m) < 2m \cdot F(n)/n$. This covers the entire computationally searched range in all hard wide families.

5.2 The Long-Rebound Lemma (Theorem D)

If $G(n) < \alpha_A/2$ and $G(m) \geq 2G(n)$, rearranging the α -start computation gives the necessary condition $\Delta \geq (nG(n) - |A|)/(\alpha_A - 2G(n))$. This eliminates short-interval rebounds.

5.3 The Run-Length Bound (Theorem E)

The interval-capacity bound $|S_A \cap I| \leq \lfloor L/a \rfloor + t \cdot \lfloor L/(N+1) \rfloor$ forces counted runs to have bounded length. For $k=2$, $t=a-1$: $R_A < 2a+4$. Combined with $F(n) \geq n/a$, this eliminates single-spike rebounds after $n > 2a^2$.

6. The Multiplication Table Connection

The counting function for one-anchor families has the form $F(x) = \lfloor x/a \rfloor + U_x$ where U_x counts distinct entries in a structured hyperbolic multiplication table $\{j \cdot (ka+i) \leq x : 1 \leq i \leq t, a \mid j\}$. This connects EP-488 to Ford's work on $H(x,y,z)$ and the generalized multiplication table problem.

6.1 Bifurcation at $t \sim \sqrt{a}$

Thin regime ($t \leq 2\sqrt{a}$): Almost collision-free. Tightest cases with ratio ≈ 1.89 . **Proved** by Theorem B.

Wide regime ($t \gg \sqrt{a}$): Heavy collisions, multiplicity ~ 1.6 . Lower ratio ≈ 1.41 . **Open**.

6.2 EP-488 as a No-Rebound Theorem

The global $\sup/\inf < 2$ is FALSE in the wide regime. But the directional statement ($m > n$) still holds because the density peak comes before the density trough. EP-488 is a no-rebound theorem: $F(x)/x$ can decrease but cannot rebound by factor 2.

7. Computational Verification

4694 one-anchor families tested (all primes $a \leq 199$, $k \in \{2,3,4\}$, all t). Zero EP-488 failures. Worst directional start always at $n = 2ka-1$. Singleton-extremal conjecture survives through $\max(A) \leq 16$.

8. The Remaining Gap

For one-anchor wide families ($t > 2\sqrt{a}$), EP-488 reduces to a near-1 one-sided proportionality theorem for $U(x) = H_B(x) - H_B(\lfloor x/a \rfloor)$ in the fixed-ratio regime. The proof must use primitivity (no element of A divides another) — a constraint that none of the 29 killed approaches exploited.

What would close it: (1) a primitivity-sensitive argument controlling collision structure, (2) Ford's proportionality constant made explicit in the fixed-ratio regime, (3) proof that the future-max envelope is maximized at $n = 2ka - 1$, or (4) application of Haddad's December 2025 Poisson–Dirichlet asymptotic.

9. Methodology

This work used a multi-model AI collaboration pipeline: Claude (Anthropic, Opus 4.6) for orchestration and strategic direction; GPT-5.4 (OpenAI, extended thinking) for deep proof search and counterexample discovery. The human operator directed all strategic decisions, including the identification of EP-488 as a multiplication table problem — the key insight that redirected the research from sieve theory to divisor-distribution theory.

29 proof approaches were explored and killed with explicit counterexamples, each narrowing the viable path. Two approaches succeeded (Theorems A and B). The adversarial structure — using one model to attack another's claims — proved essential for maintaining rigor.

10. References

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